

# Project Management System (PMS)

Java Web Application — Technical Abstract

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## Abstract

The **Project Management System (PMS)** is a full-stack, Java-based web application designed to streamline collaborative project planning, task management, team coordination, and progress reporting within an organisation. Built upon the **Java Servlet and JSP (JavaServer Pages)** technology stack, the system follows the classic **MVC (Model-View-Controller)** architectural pattern and is deployed on an **Apache Tomcat** server with a **MySQL** relational database as its persistent data store.

PMS provides role-based access control, distinguishing between **Administrators**, **Managers**, and **Developers**, each with tailored views and permissions. Administrators govern user accounts and system-wide settings; Managers oversee multiple projects, assign tasks, and schedule meetings; Developers track their assigned tasks and update their progress in real time. The application features a dynamic **Dashboard** aggregating project KPIs, an interactive **Calendar** for meeting and holiday visibility, a **Task Management** module with priority and status tracking, and a **Reports** module powered by Chart.js visualisations for data-driven decision making.

All business logic is encapsulated within dedicated **DAO (Data Access Object)** classes that interact with the MySQL database via JDBC, ensuring clean separation between the presentation, business, and persistence layers. Session-based authentication with servlet filters secures all protected resources. The project is packaged as a standard Java web application and can be compiled and deployed using a straightforward PowerShell-based workflow, making it easy to set up on any Windows development machine.

## Key Features

- **Role-Based Access Control** — Admin, Manager, and Developer roles with appropriate permissions.
- **Project Management** — Create, update, and track projects with status (Planning, In Progress, Completed).
- **Task Management** — Assign tasks with priority (Low/Medium/High) and track status (Todo/In Progress/Done).
- **Interactive Calendar** — View meetings and public/internal holidays in a monthly calendar view.
- **Meeting Scheduler** — Schedule project meetings with location, date/time, and participant visibility.

- **File Management** — Upload and attach files/documents to projects for centralised storage.
- **Reports & Analytics** — Chart.js-powered visual reports on project and task statistics.
- **Secure Authentication** — Session-based login/logout with AuthFilter protecting all resources.

## Technology Stack

| Layer          | Technology                                      |
|----------------|-------------------------------------------------|
| Frontend       | JSP (JavaServer Pages), HTML5, CSS3, JavaScript |
| Charts & UI    | Chart.js, Custom CSS                            |
| Backend        | Java Servlets (Jakarta EE / javax.servlet)      |
| Architecture   | MVC Pattern with DAO Layer                      |
| Database       | MySQL 8.0 (pms_db)                              |
| Connectivity   | JDBC via MySQL Connector/J 9.7.0                |
| Server         | Apache Tomcat 9.x                               |
| Build & Deploy | PowerShell + javac CLI                          |
| JDK Version    | Java 26.0.1+                                    |

## Entity-Relationship (ER) Diagram

The following ER diagram illustrates the database schema for the PMS, showing all entities, their attributes, and the relationships between them.





